Eliminative Information in a Generalized Monty Hall Problem IV

Multiple Prizes with Unequal Rewards

Abstract

Parts I–III developed the generalized Monty Hall problem from identical prizes, to variable reveal costs, to heterogeneous rewards and labeled priors. This note isolates the intermediate case of multiple prizes with unequal values. There are K doors, exactly $m \geq 1$ prize doors, and prize values $v_1, \ldots, v_m$, which need not be equal. We first study the exchangeable model in which the multiset $\{v_1, \ldots, v_m, 0, \ldots, 0\}$ is placed uniformly at random among door labels. In that case, unequal rewards still collapse to a single scalar, the total reward mass

$$V = \sum_{i=1}^{m} v_i.$$

Under uniform switching after t zero reveals, expected stay value is V/K and expected switch value is $V(K-1)/[K(K-1-t)]$. Thus unequal prizes alone do not destroy the one-dimensional structure. The real complexity appears only when door labels carry prior information or when switching ceases to be uniform. We formulate the exchangeable theorem, derive the constant-cost threshold, and supplement the theorem with empirical Stage 1 evidence: theory versus simulation curves and collapse plots across distinct unequal prize vectors with the same total reward mass.

1 Setup

There are K doors, with $1 \leq m < K$ prize doors. Prize door i carries value $v_i > 0$, and the remaining $K - m$ doors carry value 0. Let

$$V = \sum_{i=1}^{m} v_i.$$

We assume throughout that Monty knows realized prize locations and reveals only zero-value doors, never opening the player's initial door.

The player:

- chooses one door initially;
- observes t zero-value reveals, with

$$0 \leq t \leq K - m - 1;$$

- then either stays, or switches to one of the remaining unopened non-initial doors.

The main question is: how much of the Part I and Part III structure survives when there are *multiple prizes with unequal values*?

2 Two Distinct Models

As in Part III, the first split is conceptual.

Exchangeable unequal prizes. The multiset

$$\{v_1, \ldots, v_m, 0, \ldots, 0\}$$

is placed uniformly at random among the K door labels. Door labels carry no prior information.

Labeled unequal prizes. Door labels have distinct priors over reward, so the identity of the initial choice and the identity of Monty's reveals matter.

This note focuses first on the exchangeable case, because it is the cleanest and pins down what unequal prizes do *not* change.

3 Exchangeable Unequal Prizes

Assume the values $v_1, \ldots, v_m$ are fixed, but their placement among the K doors is uniform over labels. The player chooses an initial door uniformly. Monty reveals t zero-value doors. The player then switches uniformly among the remaining unopened non-initial doors.

Theorem 1 (Collapse to total reward mass)**.** *Under exchangeable placement and uniform switching,*

$$\mathbb{E}[\textit{stay reward}] = \frac{V}{K}$$

and

$$\mathbb{E}[\textit{switch reward after } t \textit{ reveals}] = \frac{V(K-1)}{K(K-1-t)}.$$

Proof. By exchangeability, every door has the same ex ante expected reward. Since the sum of rewards over all doors is V, the expected reward of the initial door is V/K.

The expected reward mass outside the initial door is therefore

$$V - \frac{V}{K} = \frac{V(K-1)}{K}.$$

Monty reveals only zero-value doors, so this expected outside reward mass is not altered by the reveals. After t reveals, the remaining non-initial closed set has size $K-1-t$, and symmetry implies that uniform switching spreads this outside reward mass evenly across those switch targets. Hence

$$\mathbb{E}[\text{switch reward after } t \text{ reveals}] = \frac{1}{K-1-t} \cdot \frac{V(K-1)}{K} = \frac{V(K-1)}{K(K-1-t)}.$$

□

Remark 1. *If all positive prizes are unit prizes, then $V = m$, and this reduces exactly to Part I:*

$$\mathbb{E}[\textit{stay}] = \frac{m}{K}, \qquad \mathbb{E}[\textit{switch after } t] = \frac{m(K-1)}{K(K-1-t)}.$$

So unequal prize values do not change the exchangeable theory except through the single scalar V.

4 Advantage of Switching

The switch advantage in the exchangeable unequal-prize model is

$$\frac{V(K-1)}{K(K-1-t)} - \frac{V}{K} = \frac{Vt}{K(K-1-t)}.$$

Thus the sign and comparative statics are exactly the same as in Part I:

- the switch advantage is nonnegative and strictly positive for $t > 0$;
- the gain from switching increases with reveal depth;
- unequal prize values merely rescale the payoff level by V.

This makes the exchangeable unequal-prize model mathematically neat, but not yet qualitatively new.

5 Constant Reveal Costs

Suppose each reveal costs a constant amount c. Then the net switch value after t reveals is

$$W(t) = \frac{V(K-1)}{K(K-1-t)} - ct.$$

Its one-step increment is

$$\Delta_t = W(t+1) - W(t) = \frac{V(K-1)}{K(K-2-t)(K-1-t)} - c.$$

Because the benefit term is increasing in t, the same endpoint logic as in Part I survives. In particular, the comparison between full reveal and no reveal is

$$\frac{V(K-1)}{Km} - c(K-m-1) \quad \text{versus} \quad \frac{V}{K}.$$

This yields the critical constant-cost threshold

$$c^* = \frac{V}{Km}.$$

Remark 2. *When all prizes are unit prizes, $V = m$, so the threshold becomes*

$$c^* = \frac{1}{K},$$

recovering Part I exactly.

6 Stage 1 Empirical Evidence

Stage 1 coding verifies that the exchangeable unequal-prize theorem is not just a symbolic rescaling. The simulation draws explicit prize vectors, places them uniformly among doors, reveals only legal zero doors, and estimates both stay and switch values from Monte Carlo runs.

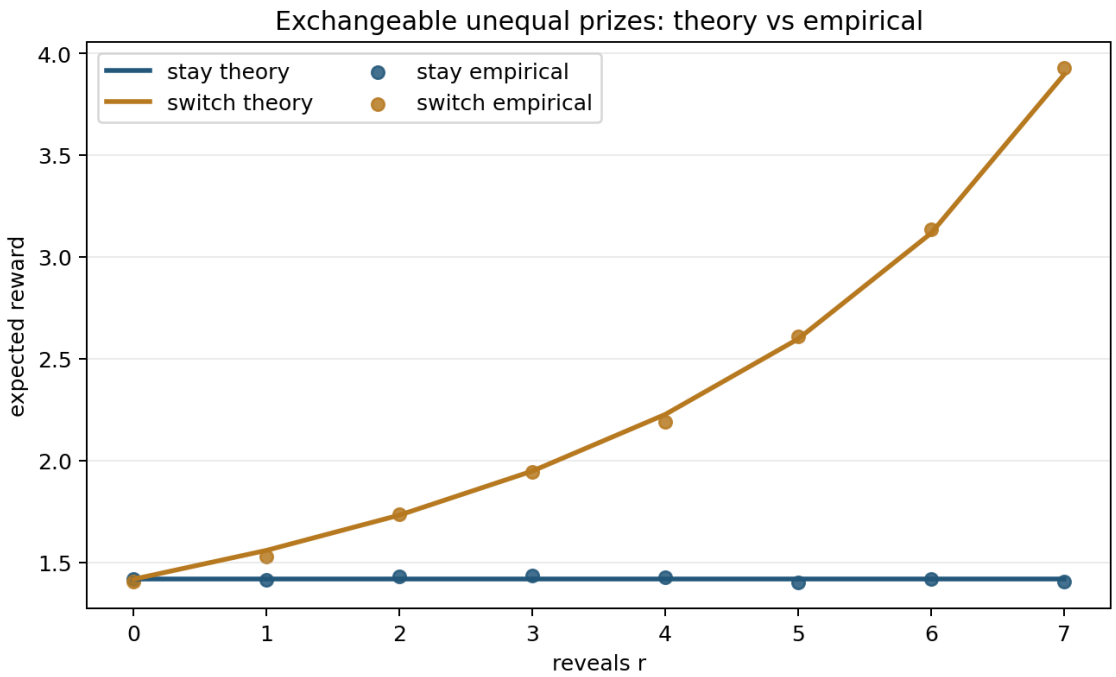

Figure 1: Stage 1 baseline verification for $K = 12$, $m = 4$, and reward vector $(1, 2, 5, 9)$. Stay value is flat at V/K, while switch value rises with reveal depth exactly as predicted by $V(K-1)/[K(K-1-t)]$. The empirical points track the theoretical curves closely.

For the baseline experiment, we fix $K = 12$, $m = 4$, and use the unequal prize vector $(1, 2, 5, 9)$, so $V = 17$. Figure 1 compares the empirical estimates against the exact formulas over all feasible reveal depths $t = 0, 1, \ldots, K - m - 1$. The agreement is close throughout. In the current Stage 1 run, the largest absolute stay error is about 0.017 and the largest absolute switch error is about 0.037, both well within ordinary Monte Carlo noise for a finite simulation of this size.

The more important question is whether the collapse to total reward mass V can be seen directly in simulation when the prize vector itself changes shape. Stage 1 therefore compares three distinct reward vectors,

$$(1, 2, 5, 9), \qquad (4, 4, 4, 5), \qquad (8, 4, 3, 2),$$

all of which satisfy $V = 17$. These vectors differ substantially in how reward mass is distributed across the $m = 4$ prize doors, but under the exchangeable model they should produce the same expected stay and switch values.

Figure 2 shows exactly that. When plotted in raw reward units, the empirical curves for same-V reward vectors lie nearly on top of one another, and they all follow the same theoretical shape. Figure 3 then divides reward by total mass V, making the scale-free claim visually explicit: after normalization, the curves collapse onto the same geometry.

These plots matter because they rule out a tempting but false intuition. One might suspect that a "spikier" prize vector such as $(1, 2, 5, 9)$ should behave differently from a flatter vector such as $(4, 4, 4, 5)$ even in the exchangeable case. Stage 1 shows that this is not so. As long as labels are uninformative and switching is uniform, the simulation respects the theorem: only total reward mass matters.

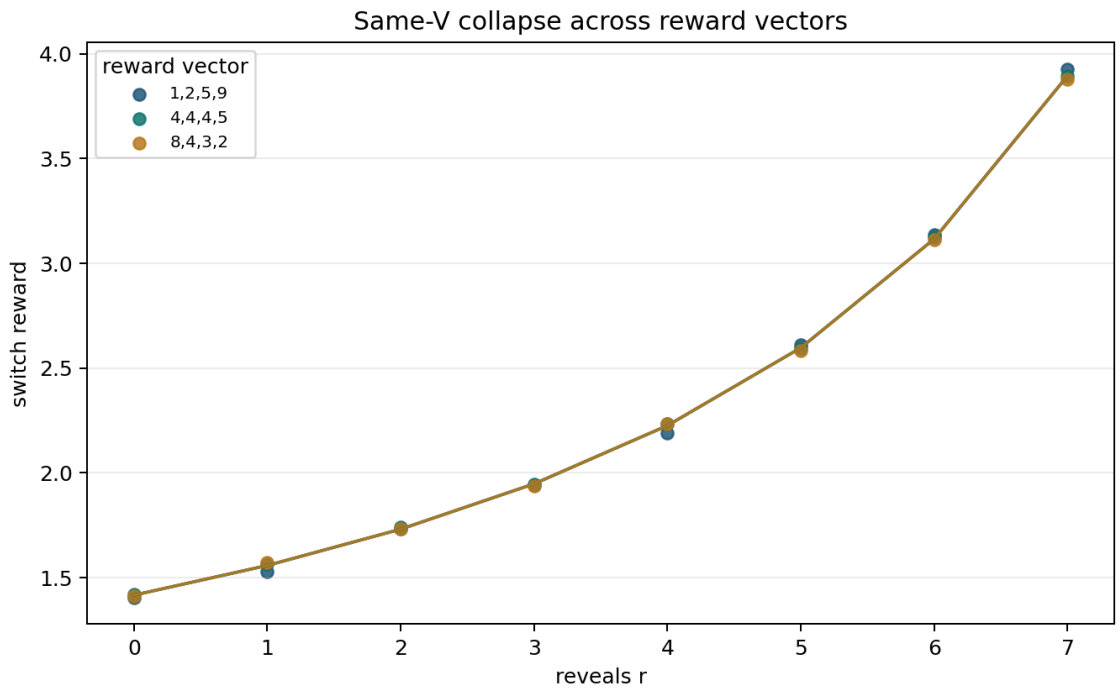

Figure 2: Same-V collapse in raw reward units. Three distinct unequal prize vectors with the same total mass $V = 17$ produce nearly identical empirical stay and switch curves across reveal depth.

7 What Unequal Prizes Do Not Yet Change

The exchangeable unequal-prize model leaves the problem essentially one-dimensional. The relevant state variable is still just the reveal count t, because label symmetry survives. The consequences are:

- stay and switch values depend only on total reward mass V;
- the exact success-law geometry from Part I survives after a scalar rescaling;
- the variable-cost analysis from Part II also carries over with m replaced by V in the relevant payoff scale;
- there is still no strategic initial choice, because all door labels are ex ante equivalent.

So unequal prizes by themselves do *not* create a new decision theory. They create a new payoff scale.

8 Where the Genuinely New Problem Begins

The interesting Part IV problem is therefore not merely "multiple prizes with unequal rewards." It is:

> multiple unequal rewards *with labeled prior information over doors.*

Once labels matter, the state ceases to be scalar. The player must track a posterior reward landscape over remaining doors. Then:

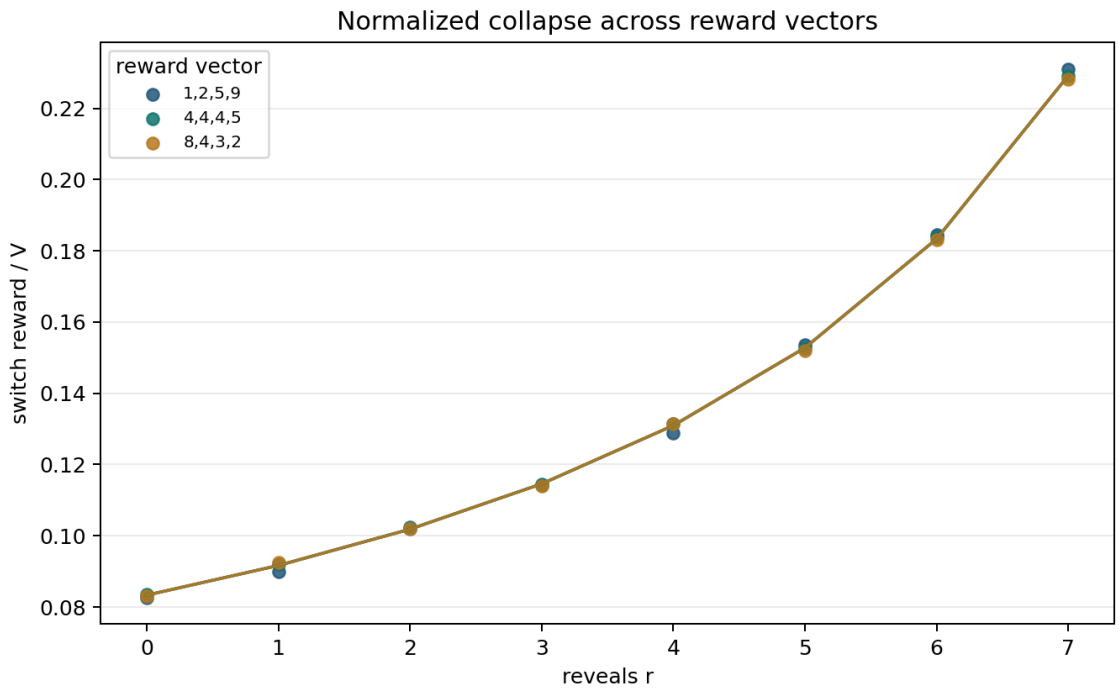

Figure 3: Normalized collapse after dividing by total reward mass V. This is the clearest empirical expression of the theorem: once values are normalized by V, the exchangeable unequal-prize model has the same reveal-depth geometry regardless of how the positive reward mass is partitioned among the m prize doors.

- the initial door can again become strategic;
- different switch rules no longer coincide;
- Monty's reveal policy can carry value-relevant information;
- multiple prize values create a richer posterior allocation problem than in Part III.

A natural first labeled model is:

$$X_j \in \{0, v_j\}$$

with

$$\mathbb{P}(X_j = v_j) = p_j, \qquad \mathbb{P}(X_j = 0) = 1 - p_j,$$

so each door has prior mean

$$\mu_j = p_j v_j.$$

This is the direct multiple-prize unequal-value analogue of the Bernoulli-value model from Part III.

9 Conjectural Part IV Questions

The next coding and theory questions should be:

1. In the labeled unequal-prize model, when does prior-best switching beat uniform switching by a large margin?

2. Can low-value initial choices preserve a better switch set when there are several high-value prize candidates rather than just one standout door?

3. How sensitive is the switch value to Monty's legal reveal policy?

4. Does a useful reduced state survive, or does the problem become genuinely high-dimensional in the posterior reward vector?

These are the questions that would make Part IV mathematically distinct from Part III.

10 Conclusion

Multiple prizes with unequal rewards split cleanly into two regimes.

In the exchangeable regime, unequal rewards collapse to total reward mass V. The theory remains scalar, and the Part I and Part II arguments survive with only a rescaling of value. The new Stage 1 evidence shows that this is not just a formal theorem: simulation produces the same reveal-depth curves for very different unequal prize vectors once total reward mass is held fixed.

In the labeled regime, unequal rewards become genuinely interesting. Then the problem is no longer about a single quantity like t or V, but about how posterior reward mass is distributed across doors, how Monty reveals zeros, and how the initial choice preserves or destroys later switching opportunities.

That labeled unequal-prize regime is the natural next coding target.